of other metabolic pathways and carbocation stability in evaluating the carcinogenic potential of NDSRIs.

Methods: We assembled structural features relevant to N-nitrosamines' carcinogenic potency. These features formed the basis for building fingerprints used in computing a quantitative relevance index and surrogate selection. Three distinct feature sets were created: basic atom types surrounding the >N-N=O group, Carcinogenic Potency Categorization Approach (CPCA) [1] derived features, and metabolic α-CH2 hydroxylation modulators from our recent research work [2]. The rationale for surrogate selection was their structural and reactivity similarity to facilitate read-across analyses for data-deficient NDSRIs.

Results: Expert visual assessment and comparison with regulatory agency surrogate choices confirmed the effectiveness of the fingerprints in identifying appropriate surrogates for NDSRI intake limit-setting. The study demonstrates that fingerprints based on identified specific structural features notably outperform traditional structural similarity methods in identifying suitable surrogate nitrosamines for carcinogenicity assessment.

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P05-27

Using expert knowledge to re-evaluate the sensitisation potential of quaternary ammonium salts

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Quaternary ammonium salts (quats) are cationic chemicals with several applications, including as surfactants, detergents, and antistatic agents. They are commonly added to cosmetics and personal care products for these useful properties and as such must be assessed for the risk of any potential toxicity after dermal exposure, including skin sensitisation and skin irritation. While experimental testing is likely to be used to inform any such a risk assessment, computational models can also form part of the overall weight of evidence. This study sought to investigate and improve the *in silico* prediction of the skin sensitisation potential of quats within the knowledge-based model Derek Nexus, based on an analysis of the existing *in vivo* data, expert knowledge about their mechanism of action, and calculations of their reactivity and lipophilicity.

A dataset of 55 quats with associated skin sensitisation data was assembled from the public domain, and the toxicity data were analysed in detail to conclude on their sensitisation and irritation potential. Expert knowledge and a thorough literature review were used to examine the skin sensitisation potential of this chemical class. Theoretical calculations were also employed, by assuming that quats sensitise via an S_N2 mechanism and calculating the reactivity and lipophilicity of a worst-case example chemical to estimate its potency. [1] Proprietary sensitisation data for 14 mixtures containing quats were also analysed to help validate the findings.

While 29 of the quats had historical positive *in vivo* data suggesting sensitisation potential, upon review these data were better explained by considering these chemicals' ability to cause skin irritancy. The proposed mechanism of action for this class involves non-covalent ion pair formation with, and subsequent disruption of, lipid membranes, which would be a very unusual mechanism to cause sensitisation but is consistent with skin irritation. These findings were corroborated by the theoretical calculation of the reactivity and lipophilicity for a worst-case C18 chain quat, which was predicted to be a non-sensitiser with a calculated EC3 value greater than 100%. This was supported by an existing QSAR for predicting the aquatic toxicity of quats which uses lipophilicity alone without the need to include a reactivity parameter, [2] again suggesting that these chemicals are not reactive toxicants. The analysis of the 14 proprietary mixtures containing quats further supported the above conclusions, as 13 of them were negative. The remaining mixture was positive; however, a database and literature search suggested that the sensitisation potential is caused by compounds other than quats in the mixture. Overall, it is likely that this class of chemicals can be skin irritants depending on dose, but they are not true skin sensitisers. The knowledge within the structural alerts in Derek Nexus has been updated to reflect this.

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P05-28

Developing an open-source harmonized Physiologically Based Kinetic (PBK) model for 4 PFAS (PFOS, PFOA, PFNA, and PFHxS) to assess the toxicological profile of forever chemicals

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Concerns regarding PFAS-induced health effects, including hepatotoxicity and immunotoxicity across various age groups, have grown steadily over time. Consequently, regulatory bodies like the European Food Safety Authority (EFSA) have continuously reduced the tolerable weekly intake (TWI) for 4 key PFAS compounds (PFOS, PFOA, PFNA, PFHxS) to the current level of 4.4 ng/kg BW/week based on immune effects in infants. Establishing these limits, often necessitates the use of physiologically based kinetic (PBK) models which leverage mechanistic information to convert blood PFAS levels into reconstructed daily exposure. Through systematic search, we found that existing PBK model for PFAS predominantly rely on the Loccisano model which was developed decades ago and lacks comprehensive mechanistic and biological perspective. Considering the current human biomonitoring data and in-vitro results, an advanced harmonized human model adhering to PBK OECD criterion for 4 priority PFAS compounds was developed. This model incorporates physiological and biochemical parameters along with mechanistic information. The liver was considered as the primary target tissue with compounds like PFOS binding to L-FABP protein with values extracted from in-vitro cell lines, defined using the Michaelis-Menten equation. Hypothetical filtrate compartment as assumed in previous models have been replaced by proximal tubule lumen and cells mediated by organic anion-transporting polypeptide (OATP) transporters, differentiated based on sex. Enterohepatic recirculation was introduced, wherein compounds are transported to intestinal tract via gastric emptying from stomach followed by transport to liver and subsequent biliary secretion back into the small intestine. Excretion occurs through feces and urine from gut and kidney respectively, with unabsorbed dose appearing in feces. Model evaluation and validation were conducted both in rats and humans at multiple doses to enhance prediction confidence. We found that compounds like PFOS and PFOA have higher renal resorption compared to other PFAS contributing to their extended half-life. Interestingly, PFHxS and PFNA exhibited sex-dependent toxicokinetics with faster elimination in females compared to male rats. This may be attributed to increased excretion through menstrual blood or reduced renal resorption. The model demonstrated good agreement with human biomonitoring data with predicted organ concentration within 2 times of observed data. The final model has been transformed into web-based user-friendly visualization platform, facilitating the application of PBK model for exposure reconstruction and estimating tissue concentration. This open-source harmonized PBK model (deepika060193/PFAS-PBPK-Model (github.com)) can serve as a backbone for regulators and other authorities to evaluate human health risk and establish regulatory guidelines.

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P05-29

Refining toxicokinetic models for enterohepatic recirculation of toxins undergoing hepatic glucuronidation

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Enterohepatic recirculation (EHR) plays a crucial role in toxicokinetics (TK), substantially influencing the internal exposure of chemicals and their glucuronidated metabolites. Apical clearance of glucuronides from hepatocytes to bile canaliculi is a critical component of this process, relying on active transport by efflux transporters. However, a notable challenge in TK modeling of EHR persists due to the lack of computational methods for predicting glucuronide clearance via the multidrug resistance-associated protein 2 (MRP2) receptor. To address this gap, we compiled a dataset containing known kinetic values concerning clearance via MRP2 of 104 different glucuronides. These measures were derived from a range of experimental settings, such as *in vivo* studies employing mutant rats exhibiting defective MRP2 expression, *in vitro* cell permeability assays, and adenosine triphosphate-dependent uptake by membrane vesicles expressing human MRP2. We standardized these values by adjusting them to account for discrepancies resulting from diverse experimental conditions. Next, we developed a Quantitative Structure-Activity Relationship (QSAR) model to predict MRP2 clearance values for glucuronides. The model was constructed based on relevant molecular descriptors and physicochemical properties from the training portion of the dataset, and subsequently validated against the test set to evaluate its robustness and predictive accuracy. Finally, we integrated the predicted MRP2 clearance values into the parameterization of human TK models. Subsequently, we compared the newly generated TK model predictions of *in vivo* systemic concentrations with non-EHR refined models and validation data to assess our method's capacity to improve the accuracy of prediction. These results offer valuable insights for understanding how metabolism impacts internal exposures and provides a useful strategy for refined simulation of enterohepatic recirculation dynamics, advancing our capacity for performing risk assessments using New Approach Methodologies.

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P05-30

The impact of notational inconsistencies in SMILES for QSAR prediction with chemical language models

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In recent years, chemical language models utilizing string representations of chemical structures, such as SMILES notation, have gained prominence in computational toxicology, leveraging natural language processing techniques. Specifically, QSAR based on chemical language models offers comparability with existing models by utilizing SMILES as input, enabling seamless integration within the field. However, despite Canonical SMILES notation being the predominant input format, the lack of standardized processing methods across generation systems introduces certain notational inconsistencies. Does this 'dialect of SMILES notation' influence the application of chemical language models in toxicology? To address this question, the present study examines the following inquiries.

Initially, we investigated the root causes of variability and discovered that many datasets contain a mix of SMILES notations with and without consideration for stereoisomerism. Among 42 datasets examined, 52.5% of compounds lacked enantiomer information that should have been assigned, whereas only 1.09% lacked cis-trans isomer information. The disparities in stereoisomer data across datasets are likely to impact the effectiveness of chemical language models. To address the impact of SMILES notation dialects, we developed two novel models tailored to handle SMILES with notational inconsistencies, and compared the following 3 models: (1) standard preprocessing without changing original SMILES notation, (2) explicit assignment of stereoisomerism information through 3D structure calculation, and (3) deliberate exclusion of stereoisomerism data. These models were then evaluated on the Ames mutagenicity test dataset, composed of 6,512 chemicals, as a case study for the QSAR based toxicity prediction. Evaluation metrics included translation performance (ability to encode/decode compound structures, [0, 1]) and prediction performance (Area Under the Receiver Operating Characteristic, AUROC of predicted values, [0, 1]). Results revealed a substantial improvement in translation accuracy with models (2) and (3) compared with model (1): (1) 0.306, (2) 0.843, and (3) 0.827. On the other hand, prediction performance exhibited slight enhancement: (1) 0.839, (2) 0.846, and (3) 0.863.From the above, we found that (1) compound databases have many notational inconsistencies, and (2) taking stereoisomerism into account would contribute to improving the performance when applying chemical language models. We are currently working on analyzing the substructures that contribute to the prediction by the attention mechanism, which learns the weights of each token in token-based data representation such as texts. It is expected that the operation of an appropriate chemical language model will contribute to the improvement of QSAR tasks in toxicology such as Ames mutagenicity prediction.

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P05-31

PBK modelling concepts for a user friendly generic bovine kinetic modelling platform to predict transfer of compounds from feed, veterinary medicines and supplements to food

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Understanding the transfer of various compounds from feed, veterinary medicines and supplements into cattle tissue or cattle-derived products such as meat and milk is essential to assess possible health risks of